Math 115A Spring 2026 Discussion 9 Handout

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1 Introduction

Today, we'll go over how to solve the midterm problems, review basic properties of functions between sets, and go over properties of linear transformations in particular.

2 Midterm Problems

2.1 Problem 1

True or false?

1. The set $M_{2\times 2}(F)$ of 2×2 matrices over a field F, with the usual addition and multiplication, is a field.
2. Let V be a vector space. For any subspace $W \subset V$, $V \setminus W$ is not a vector space.
3. Let V be a vector space and $\{u, v, w\} \subset V$. If $\{u, v, w\}$ is linearly dependent, then $w \in \mathrm{Span}(u, v)$.
4. Let V be an n-dimensional vector space. If a subset $S \subset V$ has exactly n elements, then S is a basis for V.
5. Let V be a vector space and W_1, W_2 two finite dimensional subspaces. If $\dim(W_1) = n_1$ and $\dim(W_2) = n_2$, then $W_1 \cap W_2$ is finite dimensional and $\dim(W_1 \cap W_2) \geq |n_1 - n_2|$.

2.1.1 Solutions

1. This is **False**; there are two reasons why $M_{2\times 2}$ is not a field. First, its multiplication is not commutative; second, not every nonzero matrix has an inverse.[1]

[1] If it had inverses for all nonzero elements but were not commutative, it would be called a *division algebra*; the most famous example of such an object is the quaternions, which play an important role in computer graphics.

2. This is **True**. In fact, $V \setminus W$ is *never* a subspace (as long as W is a subspace) because it does not contain the zero element.

3. This is **False** (although it closely resembles a result we've proven in this course); the reason why becomes clear if you try to prove it. Since $\{u, v, w\}$ is linearly dependent, we can find a linear combination $au + bv + cw = 0$. We'd like to say that $-\frac{a}{c}u - \frac{b}{c}v = w$, but we can only do this if $c \neq 0$. So the result fails if $\{u, v\}$ is also linearly dependent. For example, consider the subset $\{(1, 0), (2, 0), (0, 1)\}$ of $\mathbb{R}^2$.

4. This is **False**; it would only be true if we knew that S were linearly independent or that it generated V. Take $\{(1, 0), (2, 0)\} \subset \mathbb{R}^2$ for an easy example - or any subset of any (finite-dimensional) vector space that includes the zero vector.

5. This is **False**. The first example that occurred to me was $W_1 = \text{Span}\left((1, 0, 0), (0, 1, 0)\right)$, $W_2 = \text{Span}\left((0, 0, 1)\right)$. Then $n_1 - n_2 = 1$, but $\dim(W_1 \cap W_2) = 0$. If you're not comfortable with the dimension of the vector space $\{0\}$ being 0, you can construct a great many examples from larger vector spaces. For instance, let $V = \text{Span}_{\mathbb{R}}(e_1, \ldots, e_9)$, let $W_1 = \text{Span}(e_1, \ldots, e_8)$, and $W_2 = \text{Span}(e_7, e_8, e_9)$. Then $W_1 \cap W_2 = \text{Span}(e_7, e_8)$, which is 2-dimensional, whereas $n_1 - n_2 = 8 - 3 = 5$. On the other hand, the inequality sometimes holds - for example, when $W_1 = W_2 = V$.

2.2 Problem 2

Let V be a vector space, let W_1, W_2 be subspaces of V, and let U be a subspace of W_2. Prove that $(U + W_1) \cap W_2 = U + (W_1 \cap W_2)$

2.2.1 Solution

Proof. The best approach to this sort of problem is to prove both inclusions.

$(U + W_1) \cap W_2 \subseteq U + (W_1 \cap W_2)$: Let $u + w \in (U + W_1) \cap W_2$, where $u \in U, w \in W_1$. If we can show that $w \in W_1 \cap W_2$, then we will have shown that $u + w \in U + (W_1 \cap W_2)$. To do so, we use the fact that $u + w \in W_2$: for some $w' \in W_2$, $u + w = w'$, so $w = w' - u$. But $u \in U \subseteq W_2$, so $w' - u \in W_2$ (by closure of W_2 under addition and subtraction). Thus, $w \in W_2$, meaning it is in $W_1 \cap W_2$ (since it is in W_1 by hypothesis); this means that $u + w \in U + (W_1 \cap W_2)$, as desired.

$U + (W_1 \cap W_2) \subseteq (U + W_1) \cap W_2$: Consider an element $u + w \in U + (W_1 \cap W_2)$, where $u \in U$ and $w \in W_1 \cap W_2$. We need to show that $u + w$ is in both $U + W_1$ and W_2. To see that it is in $U + W_1$, observe that $w \in W_1 \cap W_2$, so $w \in W_1$; thus $u + w \in U + W_1$ by the definition of the sum of subspaces. To see that it is in W_2, we note that $u \in U \subseteq W_2$, and $w \in W_2$, so $u + w \in W_2$ by closure. Since $u + w \in U + W_1$ and $u + w \in W_2$, we have $u + w \in (U + W_1) \cap W_2$, as we wanted.

Having shown both inclusions, we conclude that the two subspaces are equal. $\square$

2.3 Problem 3

Prove that the $\mathbb{R}$-vector space $C([0,1]) = \{f : [0,1] \to \mathbb{R} \mid f \text{ is continuous}\}$ is infinite-dimensional.

2.3.1 Solution

Proof. This problem is quite tricky - especially if you don't start from the right definition of "infinite-dimensional." Recall:

Definition: A vector space V is finite-dimensional if for some natural number n, V has a basis of size n. We say V is *infinite-dimensional* if it is not finite-dimensional.

So, we need to show that $\dim(V) \neq n$ for any natural number n. To do so, we will suppose, for the sake of a contradiction, that $\dim(V) = n$. That is, V has a basis B of size n; the Steinitz Exchange Lemma implies that any linearly independent subset $L \subset V$ has $|L| \leq |B|$. To obtain our contradiction, we will find a linearly independent set of size $n+1$.

Let $L_{n+1} = \{1, x, \ldots, x^n\}$; we claim that L_{n+1} is linearly independent. For consider a linear combination $\sum_{0 \leq k \leq n} a_k x^k = 0$. The left hand side is a polynomial $p(x)$ of degree at most n, and the Fundamental Theorem of Algebra tells us that $p(x)$ can have at most n roots, unless it is zero. On the other hand, the right hand side has infinitely many roots; so $\sum_{0 \leq k \leq n} a_k x^k$ must be the zero polynomial, which means $a_k = 0$ for all k. So, L_{n+1} is linearly independent and $|L_{n+1}| \geq |B|$, a contradiction. We conclude that V cannot have had a finite basis in the first place, so it must be infinite-dimensional.

A similar approach would be to cite the last problem on Homework 3: a vector space is infinite-dimensional if and only if, for each $n \in \mathbb{N}$, it has a linearly independent subset of size n. In this version of the argument, our linearly independent subset would be be $L_n = \{1, x, \ldots, x^{n-1}\}$.

Here's another approach, which nobody actually tried, but which I think is quite elegant. Suppose V had a finite basis $B = \{f_1, \ldots, f_n\}$. Then for any continuous function $g : [0,1] \to \mathbb{R}$, we would be able to find real numbers $a_1, \ldots, a_n$ such that $\sum_{0 \leq i \leq n} a_i f_i = g$. Now, we know that two functions $[0,1] \to \mathbb{R}$ are equal iff they take the same value at each point. We therefore know that

$$\begin{aligned}
\sum_{0 \leq i \leq n} a_i f_i(0) &= g(0) \\
\sum_{0 \leq i \leq n} a_i f_i(1/n) &= g(1/n) \\
&\vdots \\
\sum_{0 \leq i \leq n} a_i f_i(1) &= g(1)
\end{aligned}$$

That is, we can restrict each function f to a function $\{0/n, 1/n, 2/n, \ldots, n/n\} \to \mathbb{R}$. Now, we have shown in class that for a finite set A, the set of functions

$A \to \mathbb{R}$ is a vector space of dimension $|A|$. So, the space W_{n+1} of functions $\{0/n, 1/n, 2/n, \ldots, n/n\} \to \mathbb{R}$ has dimension $n+1$, meaning $f_1, \ldots, f_n$ (restricted to this set) cannot span W_{n+1}. On the other hand, it is clear that each function $\bar{g} \in W_{n+1}$ can be extended to a continuous function $g : [0,1] \to \mathbb{R}$ by "connecting the dots." (We can even describe g algebraically as a piecewise function). So, take some $\bar{g} \in W_{n+1} \setminus \mathrm{Span}(\{f_1, \ldots, f_n\})$, extend it to $g : [0,1] \to \mathbb{R}$, and observe that $g \notin \mathrm{Span}(\{f_1, \ldots, f_n\}) \subset C([0,1])$. Thus, $f_1, \ldots, f_n$ does not span $C([0,1])$, and therefore cannot be a basis after all. $\square$

Several people tried to show that no finite set could span $C([0,1])$ using the following sort of argument: let $f_1, \ldots, f_n$ be the proposed basis and set $g = \sum_i e^{f_i}$ (or some other non-linear combination of the f_i's); then g cannot be in the span of the f_i's. This is the right sort of argument, and seems convincing enough, but proving that g is not in the span of $\{f_1, \ldots, f_n\}$ is much harder than it looks - partly because it's not actually true! In particular, this sort of argument fails if our basis consists of just one function, the constant function $f(x) = 1$. Then $e^{f(x)}$ is the constant function $g(x) = e$, and $g(x) = ef(x)$. (Similar arguments have similar pitfalls). If you insist that none of the functions are constant, then the claim might be true, but I'm not sure how one would prove it (and I suspect you can find more complicated counterexamples).

3 Injective, Surjective, and Bijective Functions (adapted from the Discussion 7 Handout)

I keep putting this on the handouts, and then running out of time in discussion to actually go over it. But now that we're actually talking about linear transformations, you'll need to be very comfortable working with functions, so I've included this section one last time.

One of the major developments in mathematics over the past century has been the realization that functions between objects are often more interesting than the objects themselves. As you're beginning to see, individual vector spaces are actually rather boring - there isn't that much you can say about them, and in some sense they're mostly the same. Functions between vector spaces (that is, linear maps) are, on the other hand, extremely rich. Since we'll be working quite extensively with functions this quarter, we should spend some time getting familiar with their basic properties.

Intuition: Let X and Y be sets. A *function*, or *map*, from X to Y, denoted $f : X \to Y$, is a correspondence which assigns to each element of X a single element of Y. You can think of this as a machine which "eats" an element of X and "spits out" an element of Y, and which always gives the same output for a given input.

Note that, unlike in computer science, we do not care about the "inner workings" of a function, and in some sense a function has no "inner workings". For us, a function is defined entirely by the outputs it assigns to each input. So,

$f(x) = |x|$ and $g(x) = \sqrt{x^2}$ give the same function $\mathbb{R} \to \mathbb{R}$, even though they are computed in different ways.

Here is the actual definition of a function, which can be a bit opaque.

Definition: A function $f : X \to Y$ is a set of pairs (x, y), with $x \in X$, $y \in Y$, such that for each $x \in X$ there is a unique $y \in Y$ for which $(x, y) \in f$.

Exercise/Challenge: Let $f : \mathbb{R} \to \mathbb{R}$ be a function. Explain how the "graph" of f is in fact the function f itself.

But the formal definition of function hardly matters, and as far as I know we won't make you use this definition in any proofs (although if Sihao says otherwise, you should believe him over me). Here are some properties we *will* use extensively.

Definition: Let $f, g : A \to B$ be two functions. We say $f = g$ if for all $a \in A$, $f(a) = g(a)$.

Challenge: Show that the above definition follows from the formal definition of a function.

Definition: Let A and B be sets. We say that a function $f : A \to B$ is injective if it sends each element of A to a distinct element of B; that is, if we have elements $a_1, a_2 \in A$ and $f(a_1) = f(a_2)$, then $a_1 = a_2$. We say that f is surjective if it can output every element of B; that is, if for every element $b \in B$, we can find an element $a \in A$ such that $f(a) = b$. We say that f is bijective if it is injective and surjective.

Exercise: Let A, B, C be sets, and let $f : A \to B$ and $g : B \to C$ be functions. We denote by $g \circ f$ the function $A \to C$ given by $(g \circ f)(a) = g(f(a))$; this operation on functions is called *composition*.

(a) Show that if f and g are both injective, then $g \circ f$ is injective.

(b) Show that if f and g are both surjective, then $g \circ f$ is surjective.

(c) Using parts (a) and (b), show that if f and g are both bijective, then $g \circ f$ is bijective.

Exercise: As above, let A, B, C be sets and $f : A \to B$, $g : B \to C$ be functions.

(a) Show that if $g \circ f$ is injective, then f is injective.

(b) Show that if $g \circ f$ is surjective, then g is surjective.

(c) Give an example of a case where $g \circ f$ is injective but g is not. (Hint: you can do this using finite sets)

(d) Give an example of a case where $g \circ f$ is surjective but f is not. (Hint: you can do this with finite sets as well).

Definition: Let A be a set. The *identity function* $1_A : A \to A$ is defined by $1_A(a) = a$. Sometimes this function is denoted by Id_A or I_A, and sometimes we will omit the subscript for convenience.

Exercise: Let A and B be sets and $f : A \to B$ a function. Show that $f \circ 1_A = f = 1_B \circ f$.

Definition: Let A and B be sets and $f : A \to B$ a function. A function $g : B \to A$ is called a *left inverse* to f if $g \circ f = 1_A$; it is called a *right inverse* if $f \circ g = 1_B$. If g is both a left inverse and a right inverse, it is called an *inverse* of f, sometimes denoted f^{-1}.

Exercise: Let A and B be sets, and $f : A \to B$ a function.

(a) Show that if f has a left inverse, then it is injective.

(b) Show that if f has a right inverse, then it is surjective.

(c) Conclude that if f has a right and left inverse, then it is bijective. (Note: this follows easily from parts (a) and (b)).

(d) If f is bijective, show that it has an inverse. (Note: this does *not* follow easily from parts (a)-(c); the proof will take several lines!)

Challenge: Let A and B be sets and $f : A \to B$ a function. Suppose that $g_1, g_2 : B \to A$ are both inverses (i.e. they are both left and right inverses) to f. Show that $g_1 = g_2$, that is, inverses are unique (and so we can meaningfully speak of "the inverse" of a function). As an extra challenge, show that left inverses (or right inverses) are not unique.

4 Linear Transformations

First, let's recall the definition of a linear transformation:

Definition: Let V and W be vector spaces over a field F. A *linear transformation* from V to W is a function $f : V \to W$ satisfying the following two conditions:

f preserves sums	For any $v, w \in V$, $f(v + w) = f(v) + f(w)$;
f preserves scalar multiplication	For any $v \in V$ and $a \in F$, $f(av) = af(v)$

In other words, a linear transformation is a function which "preserves" the addition and scalar multiplication of V.

Remark: You'll probably hear me use other words to describe linear transformations - most frequently, I'll call them "maps," and sometimes I might use the scary-looking word "homomorphism." These are much easier to say (and especially, to write) that "linear transformation;" they are also used in many other fields of mathematics, to describe functions with a similar feel to linear transformations.[2]

Exercise: Prove that a function $f : V \to W$ is a linear transformation (according to the definition I gave above) if and only if, for all $v, w \in V$ and $a, b \in F$, we have $f(av + bw) = af(v) + bf(w)$.

[2] To be more precise, if X and Y are sets with some extra structure, a "map" or "homomorphism" $f : X \to Y$ is a function which somehow "respects" or "preserves" this extra structure. For example, see if you can guess what a homomorphism of fields might look like.

The condition in the above exercise is a bit more "elegant" than the two conditions I gave in my definition, but I think combining the two properties of a linear transformation into a single equation makes it harder to see what's going on.

Let's see some examples and non-examples.

Exercise: Prove that the following are linear transformations:

1. The function $f : \mathbb{R} \to \mathbb{R}$ given by $f(x) = 5x$.
2. The function $g : \mathbb{R}^2 \to \mathbb{R}^2$ given by $g(\left(\begin{smallmatrix} a \\ b \end{smallmatrix}\right)) = \left(\begin{smallmatrix} 2a+b \\ 3b \end{smallmatrix}\right)$.
3. The function $h : \mathbb{R}^3 \to \mathbb{R}$ given by $h\left(\begin{smallmatrix} x \\ y \\ z \end{smallmatrix}\right) = 8x + 2y - 11z$.
4. The function $c : \mathbb{C} \to \mathbb{C}$, viewed as a function between $\mathbb{R}$-vector spaces, that sends a complex number $z = a + bi$ to its complex conjugate $a - bi$. Is it a linear transformation between $\mathbb{C}$-vector spaces?
5. The function $d : P_n(F) \to P_{n-1}(F)$ that sends a polynomial $p(x)$ to its derivative $p'(x)$.
6. The function $I : C([0,1]) \to C([0,1])$ that sends a function $f(x)$ to the function $If(x) = \int_0^x f(t)dt$.
7. Let V be the vector space of sequences of real numbers; show that the function $T : V \to V$ sending a sequence $(a_0, a_1, a_2, \dots)$ to the sequence $(a_1, a_2, \dots)$ is linear. Then show that the function $S : V \to V$ sending $(a_0, a_1, a_2, \dots)$ to $(0, a_0, a_1, a_2, \dots)$ is.

Exercise: Show that the following are NOT linear transformations

1. The function $F : \mathbb{R} \to \mathbb{R}$ given by $F(x) = 3x + 5$
2. The function $G : \mathbb{R}^2 \to \mathbb{R}^2$ given by $G\left(\begin{smallmatrix} a \\ b \end{smallmatrix}\right) = \left(\begin{smallmatrix} 2a \\ b^2 \end{smallmatrix}\right)$

Exercise: Let $T : V \to W$ be a linear transformation.

(a) Suppose that $L = \{v_1, \dots, v_n\}$ is a linearly independent subset of V. Show that if T is injective, then $\{Tv_1, \dots, Tv_n\}$ is a linearly independent subset of W.

(b) Suppose that $G = \{e_1, \dots, e_n\}$ is a generating set for V. Show that if T is surjective, then $\{Te_1, \dots, Te_n\}$ is a generating set for W.

(c) Conclude that if $B = \{b_1, \dots, b_n\}$ is a basis for V and T is bijective, then $\{Tb_1, \dots, Tb_n\}$ is a basis for W.

Challenge: Find a counterexample to (1) when T is not injective; find a counterexample to (2) when T is not surjective.

Exercise: Let $T : V \to W$ be a linear function. Let $B = \{v_1, \dots, v_n\}$ be a basis for V, and suppose that $\{Tv_1, \dots, Tv_n\}$ is a basis for W. Show that T is bijective.